

**“An Analysis of Secondary School Teachers’ Attitude and Awareness
toward the Use of Information and Communication Technology
In Relation to Teaching Effectiveness”**

A Proposal

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Title -

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1. Introduction-

Education is a process of acquiring information, knowledge, skills, values, and attitudes that enable individuals to participate effectively in society and contribute to its development. Education is essential for human development because it empowers individuals to enhance their well-being, improve their capabilities, and expand their growth opportunities. Education also fosters social participation, cultural diversity, democratic citizenship among people, and global awareness. Education is a lifelong endeavour that requires constant adaptation and innovation to meet the changing needs and demands of learners and society.

Therefore, knowledge, skills, and expertise are necessary to keep pace with this ever-changing technology and to support our educational growth, as the global educational scenario of the 21st century can thrive in an environment that emphasises creativity, imagination, critical thinking, and problem-solving skills. So, education and technology are two interrelated fields that have a significant impact on the development of society and individuals. Education is the process of acquiring awareness, knowledge, skills, values, and attitudes that enable people to learn and grow throughout their lives. Whereas Technology is the application of scientific and technical knowledge to create tools, systems, and devices that solve problems and improve human capabilities. Together, education and technology can enhance the quality and accessibility of learning opportunities, foster innovation, creativity, and prepare learners for challenges and opportunities.

The COVID-19 pandemic brought an opportunity to accelerate the adoption of digital technologies in every aspect of human life. In education, schools and universities shifted to remote learning, online collaboration and video conferencing tools, and digital assessment tools became essential for delivering education remotely. The use of ICT in education continues to evolve, with a focus on promoting inclusive and equitable access to quality education. The integration of emerging technologies and the exploration of innovative pedagogical approaches are expected to shape the future of ICT in education.

Information and communication technology (ICT) is a term that encompasses the various devices, systems, applications, and services that enable the creation, transmission, processing, and storage of digital data. ICT plays a vital role in the modern world, as it supports various aspects of human activity, such as education, health, business, governance, entertainment, and social interaction. ICT can enhance the quality, efficiency, and effectiveness of these activities, as well as create new opportunities for innovation, collaboration, and development. ICT can also contribute to the achievement of the Sustainable Development Goals (SDGs), which are a set of global objectives to end poverty, protect the planet, and ensure peace and prosperity for all by 2030.

Information and Communication Technology (ICT) has become, within a very short time, one of the basic building blocks of modern society. Many countries now regard understanding of ICT and mastering the basic skills as part of the core of education, alongside reading, writing and numeracy. The recent efforts of the Government of India (GOI) seek to deepen the use of ICT in almost every sphere of life. The Digital India Campaign (2015) strives to transform India into a digitally empowered society and knowledge economy by focusing on the three vision areas: Digital Infrastructure as Core Utility to Every Citizen, e-Governance and Services on Demand, and Digital literacy and empowerment of citizens. The three cardinal principles of the draft New National Education Policy (2016), viz., access, equity, and quality, could be served well by harnessing the huge potential of ICT.¹

Given the explosive pace of technological development, allied with the sheer creativity of tech-savvy teachers and entrepreneurs, including student entrepreneurs, it is certain that technology will impact education in multiple ways, only some of which can be foreseen at present. New technologies involving artificial intelligence, machine learning, blockchains, smart boards, handheld computing devices, adaptive computer testing for student development and other forms of education software and hardware will not just change what students learn in the classroom, but also how they learn, and thus these areas and beyond will require extensive research both on the technological as well as educational fronts.²

ICT can enable people to access diverse and reliable sources of information and knowledge, such as online libraries, databases, e-books, podcasts, and webinars. This can help people to learn new skills, acquire new perspectives, solve problems, and make informed decisions. It can also facilitate communication and collaboration among individuals, groups, and organisations across different locations, time zones, and cultures. This can foster social inclusion, cultural diversity, civic engagement, and global citizenship.

ICT is a key enabler of human progress and development in this modern era. Information and communication technology (ICT) has become an integral part of education in the 21st century. It offers various benefits for teachers and students, such as enhancing learning outcomes, facilitating collaboration, and promoting digital literacy. However, the effective integration of ICT in teaching requires not only adequate resources and skills but also positive attitudes and awareness of teachers. A teacher's positive attitude and informed awareness are pivotal in leveraging information and communication technology to create engaging, inclusive, and impactful learning experiences. At the same time, he ensures technology enhances education, fostering curiosity, equity, and digital responsibilities in students. So, **this study aims to explore the attitude and awareness of teachers toward the use of ICT in their effective teaching.**

1.1 Origin of the study –

Poor quality of education is leading to poor learning outcomes in India, ultimately pushing children out of the education system and leaving them unemployed, vulnerable to child labour, abuse and violence.³ According to the educational report, the advantages of ICT in this new teaching and learning environment, ICT offers numerous advantages and provides opportunities for:

1. Facilitating learning for children who have different learning styles and abilities, including slow and average learners, mentally and physically challenged children and those living on a low wage in isolated rural and urban areas.
2. Making learning more effective, involving more senses in a multimedia context and more connections in a hypermedia context.
3. Providing a broader international context for approaching problems, as well as having a more sensitive response to local needs, technological resources and knowledge.

1.2 Need and importance of the study

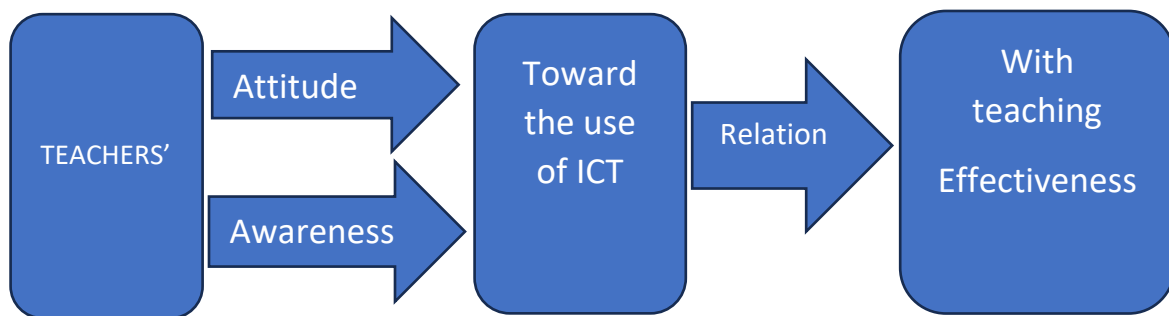
People in the present society have all shifted from old means of communication to new modes that are advanced, faster, cost-saving, and easier to use. ICT has made it feasible for people to interact globally through various connections. Technology is advancing day by day and has a major role in giving a boost to the country's economy, as it is generating a lot of employment and making it easier for people to do things efficiently and in no time. In the context of education, ICT is widely used by students for e-learning. Students take help from various websites and applications that are available for clearing their concepts and enhancing their skills through animations and other resources. Effective teachers try to ensure good student outcomes through high-quality teaching. They use all the elements of high-quality teaching to create personalised and relevant learning opportunities.

1.3 Statement of the study –

‘An Analysis of Secondary School Teachers’ Attitude and Awareness toward the Use of Information and Communication Technology in Relation to Teaching Effectiveness’. In this study, an attempt is being made to understand and implement.

1.4 Objectives of the study -

- To find out the attitude of teachers toward the use of ICT.
- To find out the awareness of teachers toward the use of ICT.
- To examine the relationship between teachers' attitudes, awareness and effective teaching.
- To compare the differences between hill area and plain area teachers' attitudes and awareness toward the use of ICT.
- To identify the barriers/challenges that hinder the integration of technology in teaching and learning.



It will also review the existing literature on the factors that influence teachers' attitudes and awareness, such as their beliefs, self-efficacy, training, and support. It will also discuss the implications of teachers' attitudes and awareness of their pedagogical practices and professional development. The study will conclude with some recommendations for improving teachers' attitudes and awareness toward ICT to enhance their teaching effectiveness. Therefore, this proposal aims to explore the following research questions:

- **What are the teachers' attitudes toward ICT in terms of their beliefs, perceptions, and emotions?**
- **What are the teachers' awareness levels of ICT in terms of their knowledge, skills, and competencies?**
- **How do the teachers' attitude and awareness of ICT influence their pedagogical practices and student outcomes?**

1.5 Hypothesis –

Based on the objectives of the study, the research hypothesised that: -

1. H₀₁: There is no significant difference in teaching effectiveness among teachers about attitudes toward the use of ICT.
2. H₀₂: There is no significant difference in teaching effectiveness among teachers concerning awareness toward the use of ICT.
3. H₀₃: There is no significant relation between attitude and awareness of teachers toward the use of ICT and teaching effectiveness.
4. H₀₄: There are no significant differences in attitude and awareness of secondary school teachers between hill area and plain area teachers.
5. H₁: There are many barriers /challenges that hinder the integration of technology in teaching and learning.

1.6 Operational definition of terms used in the research.

Attitude: - Attitude is a feeling or opinion about something or someone. It involves a person's mindset, outlook and feelings. Attitude is an individual's predisposed state of mind regarding a value, and it is precipitated through a responsive expression towards oneself. Attitudes are complex and acquired states throughout life experiences.

Psychological factors, including ideas, values, beliefs, and perceptions, determine a person's attitude. All these have a complex role in contributing to a person's attitude. An attitude can be a positive, negative, or mixed evaluation of an object, expressed at some level of intensity. It is an expression of a favourable or unfavourable evaluation of a person, place, thing, or event.

Awareness: - Awareness is a state or quality of being conscious of something. It is the ability to perceive and respond to a situation or circumstance. Awareness can also mean the knowledge or understanding of a particular subject, issue, or situation. It is often considered a key component of learning, cognition, and mindfulness.

Information and communications technology (ICT):- Information and communications technology is an integration of telecommunications and computers, as well as necessary enterprise software, middleware, storage and audiovisual, that enable users to access, store, transmit, understand and manipulate information. Information and Communication are a diverse set of

technological tools and resources, both in hardware and software forms, which are used to communicate and to create, disseminate, store and manage information* (Sanjoy Jha, Kanishka Publication, 2017-18).

Teaching effectiveness: In this study, teaching effectiveness refers to the extent to which a teacher successfully achieves the desired learning outcomes in students. It focuses on measurable results, such as improved student performance, increased active participation, enhanced motivation, and skill development. Teaching effectiveness depends on several factors, including the teacher's subject knowledge, pedagogical skills, communication abilities, positive attitude, and use of innovative strategies like ICT. It is often evaluated through academic results, student feedback, classroom observations, and peer reviews. Teaching effectiveness highlights the impact of that teaching on student learning and overall growth.

2. Review of related literature

A review of literature is a systematic and comprehensive analysis of the existing scholarly works on a specific topic or research question. It aims to identify, evaluate, synthesise, and compare relevant studies and theories, as well as to highlight the gaps, limitations, and controversies in the field. A review of the literature provides a solid foundation for developing a research problem, hypothesis, methodology, and framework. It also demonstrates the researcher's familiarity and expertise with the subject matter and their ability to critically appraise and integrate the sources.

The main purpose of this study is to investigate the teachers' attitude and awareness towards the use of information and communication technology (ICT) in their teaching practices. The study also aims to examine the relationship between these two factors and the effectiveness of teaching. Moreover, the study intends to compare the differences in attitude and awareness between teachers from hill area and plain area schools. Finally, the study seeks to identify the barriers or challenges that hinder the integration of technology in teaching and learning. so, the studies related to the topic are described separately in India in abroad and the literature review.

2.1 Studies in India

Amardeep and Singh (1998) conducted a study on the awareness of teachers of Educational Technology with a sample of forestry Institutions of U.P. The objective of the study was to assess the level of awareness of teachers about instructional materials, devices and methods. For data collection, an interview schedule and a questionnaire were used. Mean and standard deviation, and total weighted score were used for the analysis of the raw data. The findings of the study revealed that in the case of overall awareness about instructional materials, the majority of teachers were moderately aware of ET. Total Weighted awareness Score (TWS) for chalkboard was found to be maximum (133), followed by overhead projectors and transparencies (131), slides and slide projectors (118), Posters (118), chart (117), Specimens (115) and model (112).

Rathod (2002) conducted a study on computer awareness of secondary school teachers. The objective of the study was to determine the awareness of secondary school teachers of computers. To study it, 133 teachers were randomly selected from thirty-one schools. A questionnaire was used for data collection, and the collected data were analysed using percentages, mean and standard deviation. The findings of the study revealed that only 11.31% of secondary teachers were aware of computers.

Wilson (2002) did a study on the digital divide: a study of teacher awareness and efforts to address the issue. The objective of the study was to point out the awareness level of teachers regarding the digital divide. For this, descriptive research was conducted. The teachers were asked to complete a response sheet that aimed to gauge their awareness of the digital divide. In the study, it was found that the teachers were intermediate users of technology. In conclusion, the investigator suggested that teachers must be provided with the necessary opportunities to acquire the professional development necessary to equip them to integrate the technology needed in their classrooms.

Sinha, Manoj Kumar (2011) studied Information and Communication Technology (ICT) awareness among University and College teachers in the North Eastern Region of India. The objectives of the study were to know the extent of ICT amongst the teachers who participated in a refresher course on Humanities conducted at Assam University, Silchar and the teachers of the same university who have not participated in that refresher course. The researcher concluded that college and university teachers have an interest and a positive attitude towards learning ICT and applying ICT for the teaching-learning process. He found that the faculty members need appropriate training and awareness programs for the effective utilisation of resources available under the UGC-INFONET Digital Library Consortium of INFLIBNET.

Law et al. (2019). Teaching effectiveness is a critical aspect of education that directly impacts student learning outcomes and academic success. The study synthesises and examines existing research to identify strategies that can enhance teaching effectiveness, including subject matter knowledge, instructional diversity, effective communication, student-centred approaches, assessment and feedback, classroom management, student engagement, professional development, and reflective practice.

2.2 International studies

Hardiman (1988) studied the attitudes of the principals of secondary schools towards microcomputers. The sample consisted of 331 principals in the state of Georgia. A two-part microcomputer assessment survey instrument was developed for this study. ANOVA, t-test and Pearson product-moment correlation were used for data analysis. It was found that the attitude of secondary school principals towards microcomputers was positive, and their attitudes were not influenced by variables like age, sex, degree level, school size, school system, experience, etc., taken in the study.

Dupagne and Krendl (1992). reviewed the literature on teachers' attitudes towards computers. The investigators observed that the literature generally demonstrated positive attitudes of teachers towards computers. However, several studies in the author's review reported that teachers shared several barriers to integrating computers into their instruction. Though teachers believed in the instructional effectiveness of computers, they were unable to make use of the technology because they had their limitations, such as time or a lack of knowledge. The primary recommendation emerging from their review of the literature was the training of the teachers: referring to the need for schools to invest time and resources in in-service teachers and arrange workshops for teachers.

Asan (2002) carried out a survey on computer technology awareness of elementary school teachers. The sample was 252 teachers working in elementary schools in Trabzon, Turkey. The responses were collected with the help of a questionnaire. The variables taken in the study were gender, place of work, teaching experience and school type. The data were collected from the sample selected using a random sampling method. The major findings of the study were that 1) only 39% of the respondents were found to be using computers, 2) 60% of the respondents reported that they were not computer users 3) the study also confirmed that technologies like overhead projectors, printer, keyboard, modem, hard disk and video camera received the highest ranking as essential items for teaching and learning of those teachers, while the technologies like compact disc, scanner, sound

card and television card were ranked the lowest and 4) there is a significant difference between male and female teachers in their familiarity of computer technologies.

Qureshi et al, (2022). Understanding the background of ICT in education will enable educators, policymakers, and researchers to navigate the evolving educational landscape and effectively utilise ICT to enhance teaching, learning, and overall educational quality (Almufarreh and Arshad, 2023). The study aims to examine the influence of ICT integration and TR on teaching effectiveness. The study enriched the literature knowledge in the field of teaching effectiveness and the factors affecting it. Indeed, the significance of ICT in university education is profound.

Ali Ateeq, Ranyia Ali, et al. (2025) study aimed to examine the impact of Information and communication technology (ICT) and faculty readiness on teaching effectiveness in universities. particularly in the Kingdom of Bahrain. This study collected data from 181 participants using a convenient online sampling technique. Researchers gathered participants who fulfilled the inclusion criteria for the study through an online platform. The collected data were then analysed using SPSS and SmartPLS4. The results revealed a significant positive effect of ICT and faculty readiness on teaching effectiveness. The study underscores the significance of both faculty readiness and ICT integration in promoting teaching effectiveness. Educational Universities should prioritise efforts to enhance faculty readiness by providing professional development opportunities and support. Additionally, investment in ICT infrastructure and the effective integration of ICT tools can contribute to improved teaching outcomes. Hence, the study is helpful to policymakers, stakeholders, researchers, and society in general to re-evaluate and strengthen teaching effectiveness.

Investigations into students' perspectives and experiences with ICT integration underscore their overwhelmingly positive perceptions (San Isidro and Cañado, 2023). Students recognise ICT as a valuable asset in their educational journeys, affording them increased access to educational resources, fostering collaboration among peers, and facilitating effective communication with instructors (Lambrev and Cruz, 2023). However, challenges such as the digital divide, technology anxiety, and information overload must be diligently addressed to ensure equitable access to ICT resources and their optimal utilisation (Magana, 2023).

Research gap

Although several studies have explored teachers' attitudes toward ICT and their general awareness of digital tools, **most existing research has examined these variables independently** rather than in an integrated manner. A considerable portion of previous literature focuses on:

- ICT usage patterns of teachers.
- Availability and accessibility of digital resources.
- Teachers' ICT competency or training needs.
- Comparative studies between rural and urban schools.
- General impact of ICT on academic achievement.

However, **limited research** has simultaneously investigated:

1. **The combined effect of attitude and awareness toward ICT,**
2. **How these two psychological variables interact, and**
3. **Their joint relationship with teaching effectiveness, specifically among secondary school teachers.**

Moreover, many past studies were conducted in higher education or primary education settings, leaving **secondary-level teachers under-researched**, despite their critical role in implementing ICT pedagogy aligned with national education policies. Additionally, previous studies often relied on self-developed tools, whereas standardised tools measuring all three variables together are rarely used.

Therefore, there exists a clear research gap in understanding:

How secondary school teachers' attitudes and awareness toward ICT collectively influence their teaching effectiveness, using standardised and psychometrically sound instruments.

So, this study aims to fill this gap by offering **an analytical, relationship-based investigation** using reliable scales to generate more precise empirical insights.

3. Research design and methodology

An Analytical Study: - An analytical study is research that analyses relationships among variables using systematic and statistical methods to draw logical conclusions. It examines relationships, causes, effects, comparisons, or correlations between variables.

Key Features of an Analytical Study

1. **Identifies relationships between variables**
 - Example: Relationship between ICT awareness and teaching effectiveness.
 2. **Examines cause-and-effect patterns**
 - Example: Whether positive attitudes toward ICT lead to better teaching effectiveness.
 3. **Uses statistical tests**
 - Correlation, t-test, ANOVA, regression, etc.
 4. **Interprets data, not just reports it**
 - Explains *why* the results occurred.
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3.1 Research design: - The research design method of the study will be a **mixed-methods approach, combining quantitative and qualitative data collection and analysis.**

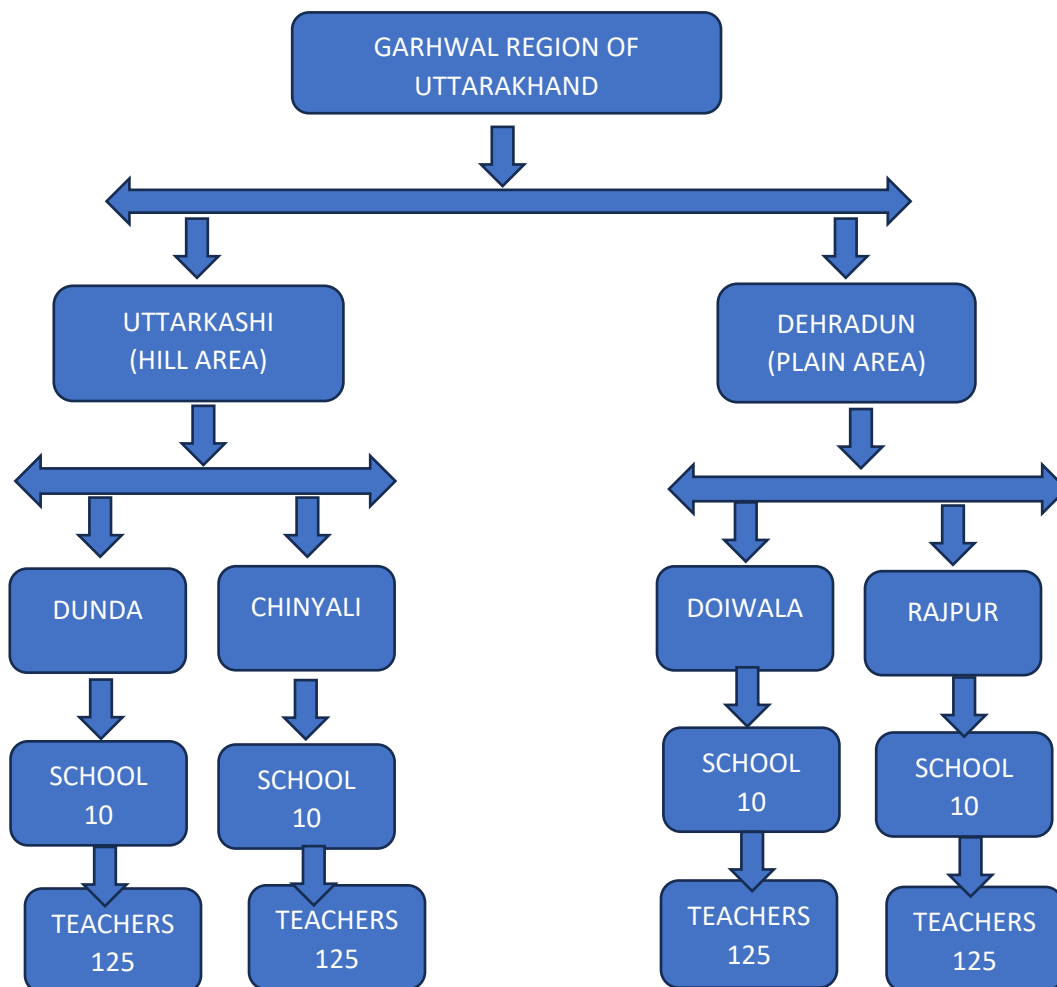
A mixed-methods research approach is a strategy that combines quantitative and qualitative data collection and analysis in a single study. This approach can enhance the validity and reliability of the findings, as well as provide a more comprehensive understanding of the research problem. A mixed-method research approach can also address different types of research questions, such as descriptive, explanatory, exploratory, or evaluative.

3.2 Method of study: - The quantitative data will consist of a **survey method** questionnaire administered to a sample of teachers from different schools and grade levels, measuring their attitude and awareness toward the use of information and communication technology (ICT) concerning effective teaching.

3.3 The population and sample: -

Population: - The population for the present study will be **Uttarakhand's secondary school teachers.**

Sampling: Multi-stage sampling is a technique that involves selecting a sample from a population in multiple steps. The first step is to divide the population into groups, known as clusters, based on specific criteria. Then, a random sample of clusters is chosen. The second step is to select a sample from each chosen cluster, either by using simple random sampling or another method. This process can be repeated for more stages if needed. The advantage of this method is that it can reduce the cost and complexity of sampling large and dispersed populations.



Research approach: - An Inductive research approach will be applied in the study.



3.4 Tools of the study: - Questionnaires, tests, and interviews will be used accordingly.

3.5 Statistics employed in the study: - MEAN, S.D. AND Correlation, t-test, ANOVA, regression.

5. Conclusion, suggestions & recommendations

5.1 Scope of the study: There is no doubt that this study will benefit the teachers, parents, students, policymakers, i.e., the government, school administrators, professional bodies, textbook writers and researchers. As for the teachers, the study will create more awareness among them towards the use of ICT facilities for teaching methods and materials in this technological age. The parents, through this study, will appreciate the usefulness of ICT to their wards and thereby create an acquisition of ICT knowledge for their educational improvement.

5.2 Delimitation of the study: There are many constraints to studying the entire population or the phenomenon, due to time and other limitations of a researcher. The study is to be delimited from various angles with the following:

1. The study is delimited to Uttarakhand's Garhwal region only.
2. The study is delimited to Uttarakhand's govt. Secondary school teachers currently teaching in the Garhwal region.
3. The study is also delimited to only 500 teachers of secondary schools.

Tentative chapter plan: -

The following chapter is planned for this proposal.

1. Introduction.
2. Literature review.
3. Research methodology.
4. Data collection and interpretation.
5. Conclusion, suggestions & recommendations / future scopes

References

The MET Project (Measures of Effective Teaching, Bill & Melinda Gates Foundation, 2012)

NCERT. (2013). ICT curriculum for the school system and teacher education. New Delhi: National Council of Educational Research and Training.

<https://ictcurriculum.gov.in>

Sharma, R. (2015). *A Study of Attitude of Secondary School Teachers Towards the Use of ICT in Teaching*. University of Delhi.

Darling-Hammond et al. (2017) – Teacher evaluation and development synthesis.

Yadav, D. (2018). *Teachers' Awareness and Use of ICT in Government Secondary Schools in Uttarakhand*. HNB Garhwal University.

Ministry of Education, Government of India. (2020). National Education Policy 2020.

https://www.education.gov.in/sites/upload_files/mhrd/files/NEP_Final_English_0.pdf

UNICEF India official website: under the Education section.

<https://www.unicef.org/india/what-we-do/education>
